ACADEMIC CENTRAL

PRACTICE EXAMINATION

NC(V) LEVEL 4

TECHNICAL PROGRAMMING

PAPER 1 · PRACTICE B

MARKS: 100 **TIME: 3 hours**

NC(V) Programming and Robotics, Level 4. Paper 1 style (theory, Python). Section A: objective questions (30) · Section B: data types and string methods, lists, tuples, dictionaries and sets, functions with default arguments and return values, file input and output, exceptions, classes and inheritance (70).

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INSTRUCTIONS AND INFORMATION

1. This question paper consists of SECTION A and SECTION B.
2. Answer ALL the questions.
3. Read ALL the questions carefully.
4. Number the answers according to the numbering system used in this question paper.
5. All program code in this paper is written in Python 3. Write your code neatly, with the correct indentation.
6. Write neatly and legibly.

SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10).

1.1.1 The data type of the value 3.75 in Python is …

A float. B int. C str. D bool.

1.1.2 The result of len("robot") is …

A 4. B 6. C 5. D "r".

1.1.3 A collection that cannot be changed after it is created is a …

A list. B tuple. C dictionary. D set.

1.1.4 The string method that changes "arm" into "ARM" is …

A lower(). B split().
C strip(). D upper().

1.1.5 A set automatically …

A removes duplicate values. B sorts text by length.
C stores key–value pairs. D keeps items in a fixed order.

1.1.6 The file mode that adds new lines to the end of an existing file is …

A "r". B "a". C "w". D "x".

1.1.7 The keyword used to handle an error without crashing the program is …

A def. B import.
C else only. D except.

1.1.8 In class Drone(Robot):, the class Robot is the …

A child class. B object.
C parent (base) class. D method.

1.1.9 The result of 17 // 5 is …

A 3. B 3.4. C 2. D 85.

1.1.10 Opening a file with the with statement …

A deletes the file afterwards. B closes the file automatically.
C makes the file read-only. D prints the file.

(10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the answer and write only 'true' or 'false' next to the question number (1.2.1–1.2.10).

1.2.1 Python lists can be changed after they are created.

1.2.2 Dictionary keys may be repeated with different values.

1.2.3 input() always returns a string.

1.2.4 A function must always have at least one parameter.

1.2.5 int("12") + 3 gives 15.

1.2.6 Opening a file with mode "w" keeps the old contents and adds to them.

1.2.7 The finally block runs whether or not an error occurred.

1.2.8 A child class can override a method of its parent class.

1.2.9 The first item of a list has index 1.

1.2.10 The __init__ method runs when a new object is created.

(10)

1.3 Choose a term from COLUMN B that matches a description in COLUMN A. Write only the letter (A–J) next to the question number (1.3.1–1.3.10).

	COLUMN A		**COLUMN B**
1.3.1	Unordered collection of unique items	A	inheritance
1.3.2	Collection of key–value pairs	B	exception
1.3.3	Error that happens while a program runs	C	set
1.3.4	Value used by a parameter when no argument is given	D	slice
1.3.5	A child class gets the attributes and methods of a parent class	E	dictionary
1.3.6	Part of a string or list, such as text[0:3]	F	casting
1.3.7	Changing a value from one data type to another	G	method
1.3.8	Function that belongs to a class	H	default argument
1.3.9	Sends a value back from a function	I	CSV file
1.3.10	Text file that stores values separated by commas	J	return

(10)

TOTAL SECTION A: [30]

SECTION B

QUESTION 2: DATA TYPES AND STRINGS

2.1 Name the data type of EACH value: (a) 25; (b) "25"; (c) 2.5; (d) True. (4)

2.2 The variable code = " arm-servo-07 " was read from a sensor log. Write the output of EACH statement: (5)

```
code = "  arm-servo-07  "
print(code.strip())               # 2.2.1
print(code.strip().upper())       # 2.2.2
print(code.strip().split("-"))    # 2.2.3
print(code.strip()[0:3])          # 2.2.4
print(code.count("o"))            # 2.2.5
```

2.3 Explain why the following code gives an error, and rewrite the second line so that it works. (3)

```
speed = input("Enter speed: ")
new_speed = speed + 10
```

2.4 Write ONE line of code that uses an f-string to print: Motor 3 runs at 120 rpm, where 3 is stored in motor and 120 in rpm. (2)

[14]

QUESTION 3: COLLECTIONS

3.1 Explain ONE difference between a list and a tuple, and give ONE example of data that should be stored in a tuple. (3)

3.2 Study the code below and write the output of EACH print statement. (6)

```
readings = [12, 15, 9, 15, 20]
readings.append(7)
print(readings)              # 3.2.1
print(max(readings))         # 3.2.2
print(readings[-2])          # 3.2.3
print(sorted(set(readings))) # 3.2.4
readings.remove(15)
print(readings)              # 3.2.5
print(len(readings))         # 3.2.6
```

3.3 A dictionary stores the battery level of each robot: (6)

```
battery = {"R1": 85, "R2": 40, "R3": 15}
```

3.3.1 Write code to change the battery level of R2 to 60. (1)

3.3.2 Write code to add a new robot R4 with a level of 100. (1)

3.3.3 Write a loop that prints the name of every robot whose battery level is below 20. (4)

[15]

QUESTION 4: FUNCTIONS, FILES AND EXCEPTIONS

4.1 Write a function called rpm_to_speed with the parameters rpm and diameter, where diameter has a default value of 0.1 (metres). The function must return the wheel speed in m/s, using speed = π × diameter × rpm ÷ 60. (5)

4.2 Write ONE line that calls the function in 4.1 for 300 rpm using the default diameter, and ONE line that calls it for 300 rpm and a diameter of 0.15. (2)

4.3 Write a program that asks the user for a number of seconds and writes the line "Delay set to … s" to the end of a file called settings.txt without deleting what is already in the file. (4)

4.4 Rewrite the code below so that it shows a friendly message if the user enters text instead of a number (ValueError) or enters 0 (ZeroDivisionError), instead of crashing. (6)

```
pulses = int(input("Pulses counted: "))
per_rev = int(input("Pulses per revolution: "))
print("Revolutions:", pulses / per_rev)
```

4.5 Explain the difference between a syntax error and a runtime error. (2)

[19]

QUESTION 5: CLASSES AND INHERITANCE

5.1 Explain the terms class and object, using a robot as an example. (4)

5.2 Write a class called Robot with: (8)

5.2.1 an __init__ method that stores name and battery (default 100); (3)

5.2.2 a method move(distance) that lowers the battery by 1 for every metre and prints the name and the new battery level; (3)

5.2.3 a method needs_charge() that returns True if the battery is below 20, otherwise False. (2)

5.3 Write a class Drone that inherits from Robot, adds an attribute altitude (starting at 0) and has a method climb(metres) that increases the altitude. Use super(). (6)

5.4 Give TWO advantages of using inheritance. (4)

[22]

TOTAL SECTION B: [70]

TOTAL: 100

END OF QUESTION PAPER